ACCESS · INDIGENOUS SKILLS 4 SUCCESS

TEACHING AND ANSWER GUIDE

Asphalt specifications

Read, calculate, check

Follow a technical table. Reconcile a delivery record. Explain a decision another person can check.

- Reading
- Writing
- Numeracy
- Digital
- Problem solving
- Communication
- Collaboration
- Adaptability
- Creativity and innovation

Classroom teaching edition · 29 October 2026
Selected reading from the historical 2006 design-build specifications.
Fictional exercise data · Technical review and classroom pilot pending.

Contents

Teach a checkable chain of evidence 3

A practical four-session route 4

Complete key · Source and table reading 4

Complete key · Four-result means 6

Complete key · Delivery and area calculations 7

Complete key · Observable communication and teamwork 8

Nine skills · Evidence that belongs to the learner 9

References and figure credits 11

Teach a checkable chain of evidence

This module develops document reading, numeracy and reliable record handling through an invented asphalt assignment. Learners must make their reasoning usable by another person. Correct arithmetic is one part of that task; exact identifiers, units, source location, version control, listening and a useful handover are equally visible parts of the workflow.

The technical reading comes from selected clauses of the **2006 Design Build Standard Specifications for Highway Construction**, adopted 20 February 2006, issued by the British Columbia Ministry of Transportation, Construction and Maintenance Branch. Keep the edition visible. It has not been established as the governing document for a present project. Do not turn the selected classroom calculations into acceptance, payment, sampling or equipment instructions. Section 501 grading work and Section 502 record work are distinct strands; a result in one is not approval under the other.

The workbook supplies all fictional inputs. Historical Table 501-E and Table 501-F values are identified separately. The source map gives clause numbers, PDF pages, printed footers and scope boundaries. A qualified technical reviewer and a classroom pilot remain appropriate next steps before operationally contextualized delivery. Do not record either review as complete unless it has occurred.

Trade outcomes and teaching scope

The trade outline's **A5, “Describe basic road and highway profiles,” task 4**, asks learners to review specifications for different types of roads. Here you practise that document-reading skill with a clearly dated B.C. specification. The unit also introduces parts of **F2, “Verify slopes, crowns and grades,” tasks 1 and 4**: units, drawing information and calculations. SP-02 and the area exercise provide introductory context for **F3, “Describe final grade preparation,” tasks 2 and 4**: fine grading and the limit of asphalt layout. A rectangle calculation does not demonstrate grade measurement, equipment calibration or finished-mat layout. Keep those practical outcomes for supervised technical training (SkilledTradesBC, 2016, printed pp. 24, 49–51; PDF pp. 26, 51–53).

Prepare the learning space

Print or provide the workbook and the selected excerpts. Supply calculators, markers, larger-print options and an agreed folder for practice files. For SP-12 provide spreadsheet software and the load register CSV. Practice the import yourself so identifiers remain text. Do not require a personal email address or real employer records. For a cohort of one, the facilitator acts as listener, checker and reviewer while keeping the learner's contributions distinct.

Invite learners to choose an accessible communication mode and explain the purpose of each observation before beginning. Read-aloud support may help a learner reach the numeracy task; document it rather than treating the result as independent reading evidence. A scribe can record a learner's message, but cannot supply its decisions and then attribute those decisions to the learner. Where a learner needs large print, provide it without asking for a diagnosis. Use the fictional setting as written; do not ask participants to represent their Nation, share community information or validate cultural content.

A practical four-session route

The following 360 minutes are a planning estimate for this expanded foundational-skills module. They are not a contractual requirement, technical-training allocation or promise that every learner will finish in that time. Extend independent practice or arrange a later digital observation when necessary.

Session	Sequence	Minutes
1 · Locate and compare	Opening and access check 10; table model 10; SP-01 12; SP-02 10; SP-03 12; SP-04 10; SP-05 11; review and feedback 15	90
2 · Mean and record	Retrieval recap 10; four-value model 10; SP-06 15; SP-07 15; SP-08 15; SP-09 15; close 10	90
3 · Calculate and revise	Opening 7; SP-10 12; SP-11 12; SP-12 25; SP-13 15; SP-14 12; close 7	90
4 · Make it usable	Opening 10; SP-15 15; SP-16 20; SP-17 12; SP-18 18; portfolio and next-step review 15	90

Begin each session with a short demonstration that exposes reasoning. For example, trace the 4.75 mm row with one finger and the Medium Asphalt Mix column with another before stating the cell. Then use a different row for the learner's response. In the mean lesson physically group four result cards, remove the oldest and add the newest. In the record lesson separate “mass reported on tickets” from “dimensions used for area” before combining them.

For SP-12, retain the formula bar or observe it live; a printed correct total alone cannot show that a learner constructed a spreadsheet formula. For SP-15–18, observe the interaction or use an agreed recording arrangement with appropriate consent and storage. A plan to communicate, collaborate or adapt is not evidence of the actual performance.

Complete key · Source and table reading

SP-01 · Leave an exact page trail

Expected address: **2006 Design Build Standard Specifications for Highway Construction; adopted 20 February 2006; Section 501, Table 501-E; PDF page 286; printed 3 of 13**. The original cover spells the month “FEBUARY”; use the normal month spelling in the learner's record without changing the stored PDF. The source publisher is the BC Ministry of Transportation, Construction and Maintenance Branch.

A valid current-project completion is: “I need to confirm the governing edition, amendments, project design and authorized review route.” Do not mark the 2006 file as current because it is the available document. If learners use the printed source card instead of navigating the actual PDF, note that distinction in the reading and digital evidence.

Figure S1 check. Learners should locate the full table title, the mm unit under Sieve Size, and the Medium Asphalt Mix heading under Percentage by Mass Passing. At 4.75 mm the Medium interval is 50–75%; the adjacent Coarse interval is 30–65%. Both are real cells, but only the requested column answers the question. Pointing at the row without identifying the column is incomplete. Offer enlargement of the original crop; do not substitute the fictional sample value for a specification limit.

SP-02 · Put work descriptions in context

For the explicitly supplied scenario: **remove existing pavement — 511.01; prepare granular base — 503.01; place and finish mix — 501.02**. These are high-level descriptions, not sufficient instructions to perform work. Actual construction sequencing and applicability depend on project design, conditions and authorized procedures. The short descriptions do not establish that all jobs include milling and granular-base preparation. Accept another explanation that correctly distinguishes the exercise's supplied sequence from a universal rule.

SP-03 · Read one cell precisely

At **4.75 mm**, the selected **Medium Asphalt Mix** interval is **50–75% by mass passing**. Both endpoints are included. The **9.50 mm** row has the **90%** upper limit. A response of “50 to 75 mm” confuses the sieve-size unit with the passing percentage. Ask the learner to label the row quantity and cell quantity separately, then try another row.

SP-04 · Convert percentage to mass

For the same fictional **1,000 g** dry sample: **90/100 × 1,000 g = 900 g** passes 12.5 mm; **5/100 × 1,000 g = 50 g** passes 0.075 mm. Estimates should be close to nine tenths and one twentieth of the sample. The percentages are cumulative, so the same smaller particles occur in multiple passing amounts. Adding the rows double-counts those portions. No actual sample handling or test method is taught.

SP-05 · Report comparisons without overclaiming

All six sample A results are within the displayed intervals. Sample B has two exceptions: **12.5 mm: 96 – 95 = 1 percentage point above** the upper limit; **4.75 mm: 50 – 49 = 1 percentage point below** the lower limit. The other four B rows are within the displayed intervals. “One percent over” is ambiguous; ask for “one percentage point above the limit.”

A suitable conclusion is: “A is within all six displayed intervals; B is outside two of them. This partial comparison does not establish full mix conformity or acceptance.” The displayed six rows omit other grading rows, properties, testing evidence and project-specific requirements. Do not reward a confident but unsupported “A is approved.”

Complete key · Four-result means

SP-06 · Move the four-result window

Tests 1–4: **(60 + 62 + 64 + 66) ÷ 4 = 252 ÷ 4 = 63%**. Difference from the 62% curve: **+1 percentage point**. Tests 2–5: **(62 + 64 + 66 + 68) ÷ 4 = 260 ÷ 4 = 65%**. Difference: **+3 percentage points**. Both differences are within ±4.5 percentage points; both means lie inside 50–75%. The first mean lies between 60 and 66; the second between 62 and 68.

A mean of 64% from all five values answers a different question. Keep four cards visible and ask which result leaves the window. Do not use individual test 5's difference from the curve to substitute for the specified four-test mean.

SP-07 · Keep both constraints visible

(49 + 49 + 50 + 50) ÷ 4 = 198 ÷ 4 = 49.5%. Difference from the 51% curve: **−1.5 percentage points**, within ±4.5. But **49.5% is 0.5 percentage point below the 50% lower limit**. A complete correction states that the curve tolerance does not waive the requirement to remain within Table 501-E's interval. See 501.15.03, PDF 286–287, printed 3–4 of 13.

This exercise establishes a failed displayed grading-interval check, not permission to change a mix design. Other rows and properties, valid test records, the governing specification and the authorized review process would be needed for an overall decision. Ask a learner who sees only one constraint to mark both on the diagram and then explain the result again in their own words.

SP-08 · Design a traceable delivery record

The form must provide: **truck number; ticket number and net weight; delivery time and station; area dimensions and material-rate calculations; notes for features such as letdowns/intersections/tapers**. It should also include the exercise's version, correction reason and checker/date fields. Those added workflow fields are ACCESS teaching choices, not a claim that the source lists them verbatim.

An identifier connects a mass to its source ticket and helps reveal omissions or duplicates. Correcting a mass must not invent another delivery. A source paragraph on PDF 299 continues as the five-item list on PDF 300; teach readers to follow the continuation. Distinguish time from station and net load mass from vehicle-plus-load gross mass. Do not introduce a tare deduction because no gross or tare inputs are supplied.

Figure S2 check. Item (d) belongs to the Road Checker’s Summary introduced at the end of PDF 299. “Include, but not be limited to” keeps the list open; a learner should not call five fields the entire specification. “Minimally … every 10 loads” states the calculation frequency. Ask the learner to identify those two qualifiers and explain them before choosing form fields. Accept a spoken, pointed or assistive explanation and record the support.

Complete key · Delivery and area calculations

SP-09 · Total and check the record

There are **10 unique tickets**, T0601–T0610. Sum: **20 + 19.5 + 20.5 + 21 + 19 + 20 + 20.5 + 19.5 + 21 + 19 = 200 t = 200,000 kg**. Estimate: 10 × about 20 t ≈ 200 t. Elapsed time **07:30–09:00 = 90 minutes = 1 hour 30 minutes**. There are nine intervals between ten time stamps. A 100-minute answer counts an extra interval.

This block is not established as a formal Lot. In 502.03.07–.08 the applicable Lot definition varies with the property assessed; the source's daily-production criteria cannot be inferred from ten deliveries. The every-ten-load record calculation and a contractual acceptance Lot are different concepts.

SP-10 · Calculate area from dimensions

Main area: **200 m × 4 m = 800 m²**. Letdown area: **20 m × 10 m = 200 m²**. Total: **800 + 200 = 1,000 m²**. The dataset specifies adjacent rectangles without overlap. Do not add lengths and call the result area. Do not measure the schematic's proportions or infer covered area from delivery station differences.

SP-11 · Calculate and explain the version 1 rate

200 t ÷ 1,000 m² = 0.2 t/m². Equivalently **200,000 kg ÷ 1,000 m² = 200 kg/m²**. This equals the fictional design target. Units are part of the answer: “0.2 kg/m²” is a thousandfold unit error.

A matching average in the exercise does not establish correct sampling, full property compliance, accepted workmanship or contractual acceptance. The target is an invented input. The source's 502.14.01 uses delivered tonnage and the applicable covered Lot area; we use the same dimensional relationship on a smaller check block without claiming its Lot status.

SP-12 · Make a digital record that can be reopened

Expected spreadsheet structure: A Ticket, B Truck, C Delivery_time, D Station, E Net_load_t; rows 2–11 contain the supplied records. `=SUM(E2:E11)` gives **200** in E13. G2 is **1000**. `=E13*1000/G2` gives **200** in H2. Ticket, truck and station fields remain readable identifiers; truck 07 should not be silently normalized to 7 when the task asks to preserve the original identifier.

Observe importing or entering, locating and entering formulas, checking units, saving in spreadsheet format, closing and reopening. Equivalent cell arrangements are valid if formulas refer to the correct data and the learner can explain them. Record actual steps and supports. A prepared teacher spreadsheet opened by the learner is evidence of opening a file, not independent formula construction. If device access fails, keep the digital status “not observed” and arrange an actual opportunity.

SP-13 · Correct the record without losing its history

Corrected mass: **200 − 20 + 18 = 198 t = 198,000 kg**. Corrected letdown: **20 m × 15 m = 300 m²**. Total area: **800 + 300 = 1,100 m²**. Revised rate: **198,000 kg ÷ 1,100 m² = 180 kg/m² = 0.18 t/m²**.

Actual rate as a percentage of the fictional target: **180 ÷ 200 × 100 = 90%**. Change from version 1: **180 − 200 = −20 kg/m²**; relative change **−20 ÷ 200 × 100 = −10%**. Less mass over more area should lower the rate, agreeing with the answer. Do not confuse the 90% target ratio with a 90% decrease, or apply historical payment bands to this incomplete classroom record.

Version 1 remains intact. Version 2 has ten rows, with T0606 corrected once. The correction log records **20 → 18 t** and **10 → 15 m** for the letdown's second dimension, plus the supplied reason. In the spreadsheet, update the existing E7 ticket row after confirming its identifier, and G2 to 1100; formulas should recalculate. Do not hard-code a correct answer over a formula.

SP-14 · Write a useful handover

One valid model:

> **Subject: Version 2 delivery record ready for review**
>
> I updated our ten-load classroom record using the correction card. Ticket T0606 now shows 18 t instead of 20 t. The letdown is 20 m by 15 m. The revised total is 198 t over 1,100 m², giving 180 kg/m², or 90% of the fictional target. Version 1 is retained. Please confirm the governing project requirements and review route before anyone uses this calculation for a real work decision.

Accept different clear wording. Look for version, changes, units, mass, area, rate and a bounded request. Do not require the model's exact sentences. Ask a peer to underline every value and trace it to the version 2 worksheet.

Complete key · Observable communication and teamwork

SP-15 · Speak, listen and check back

A successful exchange communicates **T0606 20 → 18 t; letdown 20×10 → 20×15 m; 198 t; 1,100 m²; 180 kg/m²**. The listener repeats the key information and asks a relevant question, such as “Does 18 t replace the earlier ticket mass?” The speaker confirms that it replaces the value and keeps ten deliveries. A different question about the area, units or version is valid.

Observe both turns. A written script shows preparation; the exchange shows receiving and responding. Do not use eye contact, accent, rapid speech or confidence as proxies for accuracy. For a cohort of one, the facilitator asks a genuine clarification and allows the learner to correct a deliberate, clearly classroom-only misstatement such as “So the area is still 1,000 m²?” Record whether the learner notices and explains the correction.

SP-16 · Reconcile three flawed handovers together

Round A: the 18 t correction was added as an eleventh row. Replace the original 20 t value: **200 − 20 + 18 = 198 t**, not 218 t. **Round B:** the 300 m² corrected letdown was omitted. **800 + 300 = 1,100 m²; 198,000 ÷ 1,100 = 180 kg/m²**. The reported 247.5 kg/m² is arithmetically correct for the wrong 800 m² denominator. **Round C: 198 t ÷ 1,100 m² = 0.18 t/m² = 180 kg/m²**; the error is the unit label.

Use a rota: learner A reads/checks/reports in rounds A/B/C; learner B checks/reports/reads; learner C reports/reads/checks. Each person performs each role once. With two learners, the facilitator takes the third role and rotates. With one, run three turns so the learner reads in A, checks in B and reports in C, then ask them to explain the other two corrections individually. A group answer cannot establish each member's contribution. Record a specific action, such as “compared T0606 with correction card,” “included letdown in denominator,” or “asked for t/m² to be converted.”

SP-17 · Adapt when the working arrangement changes

One valid response is to enlarge the version 2 sheet, emphasize the version and unit labels, and use it for a spoken handover. Another is to prepare a large-letter summary with the mass, area, rate and correction trail, while keeping the detailed record available. The learner must preserve the verified values and ask the reviewer to locate the rate and corrected ticket. Agree an alternative accessible format with the reviewer when needed.

Observe the change actually carried out, the feedback and any adjustment. A plan alone is insufficient evidence of adaptation. Five minutes is a scenario constraint for selecting a workable response, not a speed-based proficiency benchmark; offer an access-related time adjustment and record it. Screen unavailability does not erase the need for an actual device task in SP-12.

SP-18 · Improve the form and test the idea

There is no single correct layout. Both initial designs must retain the required information. A valid solution could place a clearly labelled “replaces” arrow between 20 t and 18 t, then show a separate totals strip **198 t / 1,100 m² / 180 kg/m²**. Another could use a two-row version table with a note that only T0606 changes. The test user should correctly identify the current rate, the corrected ticket and that 18 t replaces the earlier value.

Look for two ideas, a reasoned selection, a test with someone else, recorded actual feedback and a considered revision or reason for keeping the design. Decorative skill is not the criterion. If the user mistakes the correction for an added load, the next version should make replacement explicit. A learner who predicts what the tester would say has not yet completed the test.

Nine skills · Evidence that belongs to the learner

Use the following map to plan observation. These are local task criteria, not an official proficiency assessment or trade certification.

Skill	Activity evidence	What to observe	Useful next step
Reading	SP-01, SP-03, SP-05, SP-07	Finds the correct edition/row/column; distinguishes a stated rule from an inference	Locate a different row and explain both units
Writing	SP-08, SP-14	Produces traceable fields and a bounded handover with complete values	Revise a message after a reader's question
Numeracy	SP-04, SP-06–07, SP-09–13	Calculates and checks means, units, area, rates and change using supplied inputs	Explain the denominator without recalculating
Digital	SP-12; digital version of SP-13	Performs actual import/entry, formulas, save and reopen; preserves identifiers	Repeat with a different file and explain the formula
Problem solving	SP-07, SP-16	Identifies which constraint or input is wrong and tests a correction	Distinguish arithmetic error from wrong input
Communication	SP-15	Sends, receives, questions and confirms a technical message	Give a new recipient the same accurate handover
Collaboration	SP-16	Performs a role, shares evidence, invites a check and supports an agreed report	Rotate roles in a new three-record problem
Adaptability	SP-13, SP-17	Acts on changed information or resources while preserving essential requirements	Respond to one further supplied access change
Creativity and innovation	SP-18	Develops alternatives, tests usability and makes a reasoned improvement	Test the revised form with another reader

Observation and feedback record

Use one row per learner and task. Suggested statuses are **demonstrated independently**, **demonstrated with recorded support**, **not yet demonstrated**, and **not observed**. “Not yet” requires an opportunity and an observed attempt; “not observed” includes an unavailable device or an activity not reached. Do not turn absence of evidence into a deficit judgement.

Learner code / task	Observable action or artifact	Support used	Status	Feedback and next opportunity

Give feedback that connects an action with its effect: “You kept the old ticket and labelled the replacement, so I could check the 198 t total.” Then identify one achievable next step. Preserve useful mistakes as learning evidence when the learner agrees; the correction trail shows reasoning and growth. Before closing, ensure every learner has a next practice opportunity and understands which classroom conclusions require a qualified project decision in real work.

References and figure credits

British Columbia Ministry of Transportation, Construction and Maintenance Branch. (2006). *2006 design build standard specifications for highway construction*. Complete supplied edition.

SkilledTradesBC. (2016). *Asphalt paving/laydown technician program outline* (August 2016). Supplied outline.

Figures S1 and S2 reproduce cropped regions of the 2006 specification. Each caption records the clause or table and both page-number systems. The five explanatory SVGs are original ACCESS IS4S curriculum illustrations (2026); their captions distinguish historical source identifiers or limits from invented practice data. Crop coordinates and integrity record.